

Module 14: Unit 4 AKTU Semester Exam PYQs and Numerical Bank

Unit 4: Transport Layer

Exam Blueprint & Model Solutions

Comprehensive Bank of AKTU Previous Year Questions, 10-Marks Derivations, Hexadecimal Header Dumps, and Mathematical Numerical Solutions.

Unit 4 Transport Layer: AKTU Semester Exam PYQ Blueprint & Formula Bank

High-Frequency 10-Marks Derivations

8. **3-Way Handshake & SYN Flooding** (10 Marks)
Derive the formula for SYN Flood attack based on the given information and QoS.
Example: If the attacker sends 100 SYN packets per second, calculate the impact on the server's performance.

Mathematical Formula Bank for Semester Exam

1. **RTT Calculation** (10 Marks)
Formula: $RTT = \frac{1}{\text{SampleRate}} \times \text{SeqNumber} - \text{Pointer} - 1$
Example: If the sample rate is 0.125 and the sequence number is 100, calculate the RTT.
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1. Exam Blueprint Matrix

Question Topic	Marks Weightage	Expected Diagram / Formula
3-Way Handshake & SYN Flooding	10 Marks	Timeline diagram + SYN Cookies formula
TCP Header Format	10 Marks	32-bit grid layout + HLEN calculation

TCP Congestion Control (AIMD)	10 Marks	Slow Start & AIMD Sawtooth curve
TCP vs UDP Comparison	10 Marks	10-point architectural table
Token Bucket Numerical	10 Marks	$T = C / (M - R)$ burst calculation